

Acoustic-Based Fault Detection for Robotic Arms

Cihun-Siyong Gong¹, Senior Member, IEEE, Chih-Hui Simon Su¹, Kuo-Wei Chao,
Cheng-Yen Lin¹, and Yu-Hua Chen¹

Abstract—With the rise of Industry 4.0 and smart manufacturing, the integration of the Industrial Internet of Things and autonomous systems has become crucial for improving productivity and operational efficiency. For robotic arms, the core of automation, fault diagnosis is essential for ensuring production line reliability and sustainable operations. Traditional diagnostic methods, particularly those based on visual sensing, face challenges such as high deployment costs, insufficient contactlessness, and enormous data volumes. Acoustic-based fault detection has become a promising technique for monitoring the condition and ensuring the reliability of robotic arms. It is especially important in industrial automation, where non-invasive and real-time diagnostics are essential. This article systematically reviews recent state-of-the-art studies on acoustic signal acquisition, preprocessing, feature extraction, and classification for robotic arm fault diagnosis. Combining advanced signal processing techniques, such as time–frequency analysis, wavelet transforms, and empirical mode decomposition, with machine learning and deep learning models, has significantly improved fault detection in terms of accuracy and robustness. This review provides a detailed discussion of current research achievements and future directions, highlighting hybrid models. By integrating acoustic-sensing-based fault diagnosis with large-scale artificial intelligence models deployed in edge computing environments, users or companies can not only leverage neural network architectures to identify abnormal sound patterns amid complex background noise accurately but also directly address key challenges such as resource constraints, real-time performance, and energy efficiency at the edge.

Index Terms—Acoustic sensing, condition monitoring, fault diagnosis, Internet of Things (IoT), machine learning (ML), robotic arm.

I. INTRODUCTION

INDUSTRIAL automation has been a rapidly growing field, where robotic arms have become an indispensable tool in a wide range of applications. From automotive assembly lines and electronics manufacturing to medical surgery and space

exploration, the precision, repeatability, and immense adaptability have significantly enhanced productivity as well as improved quality control. As these robotic systems become more complex and important for mission-critical operations, ensuring their reliability and minimizing downtime have become most important issues. One of them in this regard is fault detection and diagnosis (FDD), which has been designed to identify and isolate failures or degradation of system components as early as possible to allow for timely maintenance and prevent catastrophic failures [1].

Traditionally, fault detection in robotic arms relies on physical sensors to monitor kinematic, dynamic, and thermal parameters, such as joint angles, motor currents, torques, and temperatures. Although effective, these methods often require invasive instrumentation, resulting in higher costs. As a result, they may not be able to detect subtle or early faults that do not immediately behave in the monitored variables [2], [3], [4]. In addition, sensor placement is constrained by mechanical design and environmental conditions, limiting the flexibility and scalability of such systems. As robotic systems, especially in the areas of collaborative and service robotics, become more modular and widely distributed, the need for nonintrusive, low-cost, and robust monitoring solutions is becoming obvious. In comparison, acoustic-based fault detection provides a promising alternative by using airborne or structure-borne sound signals generated during the operation of robotic systems. These acoustic emissions (AEs) can be derived from various mechanical interactions such as friction, impacts, wear, loosening components, and electrical activities in motors or actuators. Acoustic sensing offers several advantages. For example, it is noncontact, does not interfere with robot motion, and can often detect faults earlier than conventional metrics, such as vibration or temperature [5]. As a result, they encode valuable information on the health of internal components and the dynamic behavior of the system [6], [7].

Acoustic-based methods have been widely explored in areas, such as rotating machinery [8], bearing and gear fault detection [9], and structural health monitoring [10], and have been demonstrated to successfully detect anomalies under variable load conditions. For example, Wang et al. [11] proved the effectiveness of acoustic sensors in identifying early-stage bearing defects in high-speed spindles, while Chen et al. [12] applied time–frequency analysis on audio signals to detect gear pitting in noisy industrial environments. However, their applications in robotic arms are still underexplored, especially under field conditions. Robotic arms bring new challenges that differ from traditional rotary machines, including multijoint kinematics, varying

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The authors are with the Department of Electrical Engineering, National Central University, Taoyuan 320317, Taiwan (e-mail: alexgong@cc.ncu.edu.tw; chihhui.su@gmail.com; eric.chao1128@gmail.com; 114521072@cc.ncu.edu.tw; 113581001@cc.ncu.edu.tw).

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payloads, various actuation types (e.g., electric, pneumatic, and hydraulic), and complex trajectories. These factors cause highly nonstationary and dynamic acoustic patterns, making signal interpretation and model generalization more complex.

The acoustic environment in industrial settings adds another layer of complexity. Robots often operate in close proximity to other machines, human operators, and reflective surfaces that create a lot of background noise and signal interference. In this case, robust signal processing and adaptive feature extraction techniques become important to isolate fault-related acoustic signatures [13]. Many techniques, such as short-time Fourier transform (STFT), wavelet transform (WT), empirical mode decomposition (EMD), and variational mode decomposition (VMD), have been developed to decompose nonstationary acoustic signals into more interpretable time–frequency components [14], [15], [16]. These approaches can help locate transients, identify fault frequencies, and extract features that are more conducive for fault classification.

At the same time, the progress in machine learning (ML) and deep learning (DL) has enabled more sophisticated data-driven models to handle the complex acoustic data. Classical algorithms, such as support vector machines (SVMs), random forests (RFs), and Gaussian mixture models (GMMs), have been used to classify faults using handmade features derived from the time, frequency, or time–frequency domains [17], [18]. Recently, deep neural networks, including convolutional neural networks (CNNs), recurrent neural networks (RNNs), autoencoders (AEs), and transformers, have already shown great potential in learning robust and hierarchical representations directly from raw or minimally processed audio inputs, often beyond traditional methods in terms of accuracy and generalization [19], [20]. These developments open the door to real-time intelligent fault monitoring systems for robotic platforms. In experimental equipment, researchers have demonstrated the feasibility of using spectrogram-based CNNs to classify robotic joint faults [21] or long short-term memory (LSTM)-based models to monitor the temporal evolution of abnormal acoustic events during robot motion [22]. These methods are particularly promising in collaborative robotic environments, where maintaining high availability and safety is important.

Although they have outstanding achievements, there are still some research gaps and challenges. Variations in robot designs, materials, actuators, and operational profiles cause different acoustic signatures, which reduce the transferability of trained models between different platforms [23]. This problem is exacerbated by the lack of standardized datasets for acoustic fault detection in robotic systems. Because existing studies are mostly confined to controlled laboratory environments and small-scale experiments, benchmarking and comparative evaluations are difficult [24]. In addition, annotating acoustic fault data is manual and prone to mistakes, especially for early or intermittent faults [25]. Overcoming these challenges requires a comprehensive understanding of the physical acoustics of robotic systems, a clear classification of failure types, and the development of scalable data-driven frameworks that can keep robust performance under real-world variability. Transfer learning, data augmentation, unsupervised learning, and domain adaptation techniques may play a crucial role in closing the gap between research prototypes and deployable systems [26].

This study provides a comprehensive overview and in-depth discussion of the current state of acoustic-based fault detection in robotic arms. It categorizes existing methods according to the types of faults they solved, the signal processing techniques, and the ML techniques. Also discuss the key datasets, evaluation metrics, and system-level considerations. By integrating previous research and pointing out some of the unsolved challenges, this study comprehensively explores feasible solutions for acoustic sensing fault diagnosis based on edge computing, providing real-time, efficient, and sustainable health monitoring capabilities for industrial robotic arms. By deploying a specially optimized neural network model, this article provides a groundbreaking application example for the practical application of large-scale artificial intelligence (AI) models in resource-constrained industrial Internet of Things (IIoT) environments. Future technological developments could further explore domain adaptation techniques to improve the model's generalization capabilities across diverse device models, or incorporate multimodal sensor data for more robust diagnosis. It aims to help future research develop acoustic fault detection systems that are not just more reliable and scalable but also more compatible with the needs of next-generation robotic arms.

In the medical domain, surgical robotic systems (such as the da Vinci system), despite their high-precision control and stability, have predominantly relied on sensory feedback and kinematic models for fault detection. Recent studies, however, have integrated DL with multimodal sensing (acoustic, force, and vision) to predict joint friction and servo-control anomalies, thereby significantly enhancing the safety of clinical operations [27], [28].

Within production line and logistics applications, prolonged high-speed operation of robotic arms frequently leads to wear in the reducers and abnormal noise from the electric motors. Corresponding research utilizes acoustic signals in conjunction with CNNs for anomaly identification, enabling the early detection of potential failures and mitigating the risk of production line downtime [29], [30].

In semiconductor manufacturing applications, robotic arms operate within vacuum and cleanroom environments, where any minute anomaly could result in process contamination. Recent literature has employed acoustic sensing combined with generative models [e.g., variational autoencoder (VAE) and generative adversarial network (GAN)] for learning nonstationary signal characteristics, which effectively improves the identification sensitivity for subtle friction and pneumatic valve irregularities [31], [32].

For precision machining and computer numerical control (CNC) systems, multi-axis synchronous motion results in highly nonstationary acoustic features. Research indicates that combining time–frequency features [e.g., wavelet packet transform (WPT) mel-frequency cepstral coefficient (MFCC)] with the Transformer model can accurately identify minor tooling anomalies, leading to a significant improvement in machining quality and reliability [33], [34].

Collectively, while fault diagnosis techniques are relatively mature in rotating machinery, the inherent characteristics of robotic arms—specifically their multiple degrees of freedom, nonstationary acoustic behavior, and task variability—render

traditional methods difficult to apply directly. The present study thus addresses these specific challenges by proposing a scalable acoustic DL diagnosis framework designed to be extensible across various application fields, thereby highlighting its innovation and necessity.

The overall arrangement of the present study is structured according to the logical framework outlined earlier. Section I establishes the research motivation and problem background, specifically highlighting the advantages of acoustic technology over vibration and current-based methods concerning noninvasiveness, real-time capability, and cost effectiveness. Section II conducts a systematic literature search and thematic categorization to analyze the principal research directions and technological context over the past decade. Sections III and IV synthesize the developmental trends in signal processing and ML models, further elaborating on the scalability of these methods for robotic arm systems. Section V integrates the identified challenges and future research directions based on the literature synthesis, with particular emphasis on offering concrete suggestions regarding the issues of insufficient datasets, model generalizability, and explainable artificial intelligence (XAI). Finally, Section VI concludes this article.

II. LITERATURE REVIEW

A. Overview of Acoustic-Based Fault Detection

Acoustic-based fault detection is getting more and more attention due to its nonintrusive nature, cost effectiveness, and capability of capturing subtle anomalies arising from mechanical wear, misalignments, and structural degradation. In contrast to conventional sensor-based methods that rely on physical measurements such as torque, current, or temperature, acoustic methods harness airborne or structure-borne sound generated during mechanical interactions within the system. These include friction, impacts, fluid flow irregularities, and joint backlash, all of which can reflect underlying faults in components such as actuators, gearboxes, joints, or end-effectors.

Applications of acoustic fault detection have been widely used in fields, such as rotating machinery [35], bearings [36], electric motors [37], and pipeline systems [38]. These areas benefit from relatively static operating conditions and cyclical mechanical modes. However, the greater variability in the robot arm's motion, multijoint actuation, and task-specific dynamics makes the application of acoustic technology more complex and difficult to apply directly. This variability causes nonstationary and context-dependent AEs, thereby demanding more sophisticated methods for signal analysis and model interpretation. For example, Qurthobi et al. [39] studied mechanical fault detection in industrial robotic arms by analyzing acoustic signals from individual joints. Their study highlighted that the multijoint movements and task-specific dynamics of robotic arms can cause nonstationary acoustic characteristics, posing important challenges for the application of conventional acoustic fault detection methods. To overcome these challenges, the researchers applied advanced signal processing and ML techniques, including neural networks, to improve the accuracy and reliability of fault diagnosis.

Although robotic arm research is still in its infancy, the field has seen significant growth due to the increasing use of collaborative robots and the need for continuous monitoring in flexible manufacturing environments. As industrial robotics moves toward smarter, data-driven acoustic fault diagnosis methods will play a key role, especially in situations where physical access or sensor integration is limited.

B. Signal Acquisition and Preprocessing

The efficacy of acoustic fault detection depends a lot on signal acquisition hardware and preprocessing methods. Microphones, ranging from omnidirectional to directional condenser types, and piezoelectric sensors are commonly used. Their placement is often optimized near potentially fault-prone areas such as joints or gearboxes to maximize the signal-to-noise ratio (SNR). Some studies have employed microphone arrays or distributed sensing architectures to improve the localization of fault sources and isolate specific components [40]. In noisy industrial environments, robust preprocessing becomes essential. Key preprocessing techniques include the following.

- 1) *Bandpass filtering*: Only keep frequency components relevant to mechanical activity while throwing away irrelevant low-frequency hum or high-frequency electronic noise.
- 2) *Spectral subtraction and adaptive filtering*: Spectral subtraction is used to suppress both stationary and nonstationary noise components. Adaptive filtering can dynamically adapt to the changing acoustic environment [41].
- 3) *Segmentation*: The signal is segmented according to the robot's motion cycle or task state, which usually uses encoder feedback or joint state data, to align sound events with the operating environment [42].
- 4) *Amplitude normalization and silence removal*: They are used to normalize input amplitude and remove low-energy segments that lack diagnostic value.

C. Feature Extraction Techniques

To interpret the sound signals, we need to transform the raw audio into a compact information representation, which is critical for both classical and deep-learning-based detection systems. Feature types are usually divided into the following three categories.

- 1) *Time-domain features*: Time-domain features include statistical measures such as root mean square (RMS), skewness, kurtosis, zero-crossing rate (ZCR), and peak-to-peak values. These features are sensitive to signal amplitude variations and impulsive noise. This condition usually indicates a mechanical shock or impact [43].
- 2) *Frequency-domain features*: Using fast Fourier transform (FFT), frequency-domain features capture spectral characteristics such as spectral centroid, bandwidth, harmonic distortion, and power spectral density (PSD). These features are effective for identifying periodic faults, such as gear mesh anomalies or motor whine [44].

3) Time–frequency representations

- a) *STFT*: For identifying transient and changing spectral content in robot motion, it provides a spectrogram representation that supports convolutional feature extraction [45].
- b) *WT*: It provides multiresolution analysis and is ideal for isolating transient faults, such as connector misalignment or brief stall events [46].
- c) *EMD and VMD*: The signal is decomposed into intrinsic mode functions (IMFs) or modes, which adaptively reveal fault-related signal components that are masked by noise [47].
- d) *Wavelet packet decomposition (WPD)*: This is an extension of the WT, which can decompose both approximate coefficients and detail coefficients, allowing for a fine-grained time–frequency analysis. The WPD has shown excellent performance in rotating machinery fault diagnosis, especially when combined with feature importance ranking methods, which can improve the accuracy of classification and reduce the redundant features [48].

The feature extraction is important for developing interpretable models and enabling knowledge transfer across various robotic platforms and tasks.

D. Classical ML Approaches

Traditional ML models are frequently applied to engineered acoustic features for fault classification or anomaly detection. Common algorithms include the following.

- 1) *SVMs*: Suitable for high-dimensional and small-sample-size problems, SVMs are often used with time–frequency features to distinguish the faulty conditions from the healthy conditions. For example, Scanlon and Bergin [47] proved that SVMs achieved a classification accuracy of 85–98% when applied to acoustic degradation analysis of rotating machinery.
- 2) *k-nearest neighbors (k-NN) and Naïve Bayes*: These models work very well for problems with simple decision boundaries, but their performance often degrades in high-noise environments or with complex signal distributions due to their limited ability to weigh or adapt to feature relevance.
- 3) *Decision trees and RFs*: These models are particularly effective in managing noisy and redundant features. Ensemble methods, such as RFs, have demonstrated strong performance in multimodal datasets that combine acoustic and vibration signals [48].
- 4) *GMMs*: GMMs are employed to capture temporal dynamics and probabilistic state transitions in sequential diagnostic tasks. Yu et al. [49] reported that GMM-based models have better performance than multilayer perceptrons in accurately identifying fault types and severities in bearing fault diagnosis.
- 5) *Hidden Markov models (HMMs)*: These are particularly effective in modeling the sequential nature of robotic operations and identifying anomalous transitions

between task phases, especially in periodic or repetitive activities [50].

Although these approaches perform well in structured circumstances, they often face difficulties in generalizing across different robotic platforms, task domains, and operating conditions. Challenges such as domain shift, data imbalance, and noise variability continue to hinder their broader applicability.

E. DL for Acoustic Fault Detection

The emergence of DL has greatly promoted the development of acoustic fault detection systems, enabling automatic feature extraction and improving scalability across different applications. Several DL architectures have been widely used in this domain.

- 1) *CNNs*: The CNN is widely used in 2-D time–frequency representations, such as mel-spectrograms and STFTs, to automatically extract spatial patterns that indicate fault conditions. This model has demonstrated high classification performance in detecting abnormal conditions in robotic actuators and gearboxes. For example, Gong et al. [51] proposed the audio spectrogram transformer (AST), which is based on the Transformer architecture while combining CNN-style feature processing. The model attained state-of-the-art performance across multiple audio classification benchmarks.
- 2) *RNNs and LSTM networks*: These architectures effectively capture temporal dependencies and signal evolution, which are critical in robotic systems with sequential motions and variable load conditions. Gao et al. proposed a hybrid DL method, combining an extended wide-first-layer CNN with LSTM for rotating machinery fault diagnosis. It shows high accuracy in a nonstationary environment [52]. In addition, Borré et al. [53] developed a machine fault detection model based on a hybrid CNN-LSTM attention mechanism, which showed higher performance in complex industrial scenarios.
- 3) *AEs and VAEs*: AEs, especially VAEs, have been widely applied to unsupervised abnormal detection by learning compact and informative latent representations of normal acoustic signals. Abnormal conditions are detected when reconstruction error surpasses a predefined threshold. For example, Singh [54] presented an integrated enhanced graph AE by using graph attention network (GAT) and Transformer-based encoder for robust fault diagnosis. This technology achieves high accuracy in identifying anomalies without requiring labeled abnormal data [54].
- 4) *Transformer-based models*: Recent research has introduced transformer architectures for acoustic event modeling. These models utilize self-attention mechanisms to capture long-range temporal dependencies, therefore eliminating the need for repetition. Zhang et al. [55] proposed the time-series vision transformer, which combines convolutional layers for local feature extraction with transformer encoders to model long-term temporal patterns, achieving exceptional diagnostic capabilities across diverse datasets.

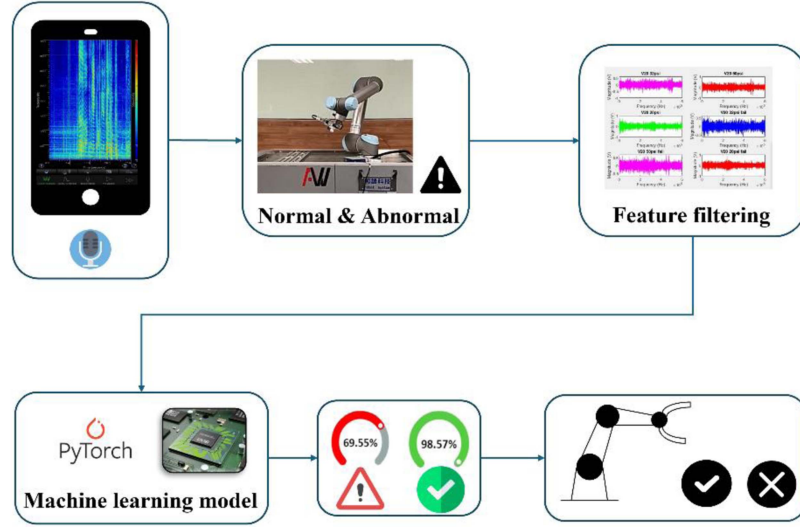


Fig. 1. Fault diagnosis process of a robotic arm using acoustic signals.

Although DL models have shown good performance that outperforms traditional methods, challenges still exist in terms of interpretability, high computational cost, and the need for a lot of training data. Current trends emphasize developing lightweight models suitable for real-time fault monitoring and integrating explainable AI methods to enhance transparency in robotic diagnostics.

F. Application to Robotic Arms

Although acoustic-based diagnostics have traditionally focused on general machinery, recent research has increasingly looked toward robotic arms, using DL and multimodal data fusion to improve fault detection capabilities. For example, Wang et al. [56] developed a pipeline robot fault diagnosis system using sound signal recognition. They used a ResNet-50 to classify different fault types with high accuracy, which shows that DL models can effectively handle complex acoustic patterns associated with robotic mechanisms, demonstrating their effectiveness. In another study, Yurdakul and Tasdemir [57] used the DL architecture DenseNet-169 to analyze acoustic signals converted into mel-spectrograms to detect faults in industrial machines. Their method achieved accuracies ranging from 97.17% to 99.87% under different noise conditions, a result that highlights the robustness of DL methods for acoustic-based fault diagnosis.

In addition to single-modal methods, multimodal sensor fusion has also been explored to improve the robustness of fault detection. Inceoglu et al. [58] applied FINO-Net, the deep multimodal sensor fusion framework that combines audio, RGB, and depth data to detect operational failures of service robots. Their model achieved 98.60% detection accuracy and 87.31% classification accuracy, which shows the benefits of integrating multiple sensory inputs [58]. Fig. 1 illustrates the acoustic-based fault detection workflow proposed in this study. The system captures the operational sound emitted by the robotic arm, processes the acquired features, and subsequently utilizes an ML

model for analysis, thereby enabling real-time fault diagnosis and condition monitoring.

Current studies have also examined robotic-specific datasets and domain knowledge. For example, Yun et al. [59] proposed an AE-based abnormal detection framework for industrial robot arms using internal sound sensors. They trained the AE under normal operating conditions and enabled it to detect exceptions through reconstruction errors, which enables faults to be identified effectively without requiring labeled fault data. In addition, Raouf et al. [60] developed a transfer-learning-based intelligent fault detection method for industrial robotic systems. By converting 1-D current signals into 2-D scalograms and utilizing a pretrained VGG16 model, their method demonstrated effective fault detection performance under variable working conditions, improving the generalization capabilities of robotic prognostic health management.

Although these advances exist, several challenges still exist when applying acoustic-based fault detection to robotic arms.

- 1) There is high variability in joint kinematics, actuation forces, and payload interactions.
- 2) There are background noise and acoustic interference from other machines or human activities.
- 3) The lack of public benchmark datasets for acoustic-based robot diagnostics hinders reproducibility and model comparison.

To address the aforementioned limitations, recent methods using digital twins for fault simulation, transfer learning for domain adaptation across robots, and self-supervised learning to reduce dependence on labeled data have been developed. These strategies aimed to improve the scalability and deployment preparation of acoustic-based fault detection in real-world robotic applications.

III. SIGNAL PROCESSING TECHNIQUES

Signal processing plays a key role in acoustic-based fault detection for robotic arms, acting as the bridge between raw often noisy audio input and informative features usable by

diagnostic models. Robotic arm operations generally involve dynamic nonstationary motion patterns, multijoint interactions, and variable load conditions, all of which result in complex acoustic signatures. To effectively isolate fault-related information from these signals, a variety of signal processing techniques are employed to improve the SNR and extract features that reflect mechanical anomalies. This section categorizes signal processing methods into four domains: time-domain, frequency-domain, time–frequency domain, and advanced feature representations.

A. Time-Domain Analysis

Time-domain signal analysis includes extracting statistical or morphological features directly from raw vibration signals. Due to its computational simplicity, this method is especially suitable for real-time applications and embedded systems, where processing power and memory are often limited [61]. Time-domain features are easy to compute and interpret, making them widely used in the condition monitoring of rotating machinery and robotic systems. However, they may be insufficient for detecting subtle or frequency-specific fault signatures and highly sensitive to noise [62]. The commonly used time-domain features include the following.

- 1) *RMS*: RMS represents the energy content of the signal and is effective for identifying faults caused by friction or impacts. It is one of the most frequently used indicators in vibration-based condition monitoring [9].
- 2) *Peak-to-peak amplitude*: It captures the full range of signal oscillation. It increases in the presence of impulsive faults such as bearing defects or gear tooth breakage, making it useful for detecting mechanical disruptions suddenly [63].
- 3) *Skewness and kurtosis*: These statistical metrics describe the asymmetry and peakedness of the signal distribution, respectively. A high kurtosis value generally indicates the presence of transient shocks, which are characteristic of early-stage bearing or gear faults [64].
- 4) *Crest factor*: This feature, defined as the ratio of peak amplitude to the RMS value, is especially effective for identifying sharp and impulsive anomalies associated with loose or damaged components. It is quite effective in distinguishing normal operational states from fault conditions [65].
- 5) *ZCR*: The ZCR measures how frequently the signal changes sign. It is sensitive to high-frequency components and can be used to detect high-frequency vibrations usually associated with incipient faults [61].

Additional time-domain features, such as the impulse factor, shape factor, and margin factor, have also been proposed to improve diagnostic performance. These features give more detailed descriptions of signal morphology and can improve fault classification accuracy in multifault scenarios [66].

Although time-domain analysis alone may not be sufficient to detect complex or frequency-sensitive failures, its simplicity and interpretability make it an essential part of hybrid diagnostic frameworks. Recent studies have promoted combining

time-domain features with frequency or time–frequency analysis to achieve more robust and accurate fault detection [62], [63].

B. Frequency-Domain Analysis

Frequency-domain analysis transforms time-series signals into the spectral domain, revealing the distribution of signal energy at various frequencies. This transformation plays a key role in identifying mechanical faults, as such anomalies often appear as spectral pattern alterations, including harmonic distortions and sideband structures. By examining these spectral characteristics, one can detect and diagnose issues, such as gear wear, bearing defects, or structural imbalances [67].

Key techniques in frequency-domain analysis are as follows.

- 1) *FFT*: The FFT breaks down the signal into its constituent sinusoidal components, promoting the identification of harmonics linked to motor imbalance or gear wear. Specifically, the FFT has been effectively used in diagnosing gear faults by revealing sideband modulations around gear meshing frequencies [67].
- 2) *PSD*: PSD represents the signal's power distribution over frequency and is commonly used to quantify energy at specific frequency bands. This metric gives insight into the severity and progression of faults over time. PSD is effective in tracking structural deterioration of steel beams and mechanical systems, where spectral energy patterns shift due to damage accumulation [68].
- 3) *Cepstral analysis*: Cepstral analysis is capable of identifying periodic structures within the frequency domain by analyzing the logarithm of the power spectrum. It is quite useful for revealing quefrency components corresponding to repetitive impacts, such as gear meshing or bearing faults. This method has demonstrated strong performance under variable operating conditions, which are typical in robotic systems [69].

Although it is interpretable and has wide applicability, traditional frequency-domain methods rely on the assumption of signal stationarity. However, robotic arms typically experience varying trajectories, velocities, and loads, which make them susceptible to transient and nonstationary signals. To address this limitation, advanced spectral features such as MFCCs have been adopted. MFCCs give compact representations of the frequency content and have shown effectiveness in early fault detection in complex mechanical systems like wind turbine gearboxes operating under variable-speed conditions [70]. Fig. 2 illustrates an AI-assisted fault diagnosis workflow for a robotic arm. This process integrates sensory signals and employs AI for feature extraction and classification, thereby enabling intelligent control.

C. Time–Frequency Analysis

Time–frequency analysis techniques address the limitations of stationary signal analysis by giving localized frequency content over time. These methods are especially effective in capturing operational state transitions (e.g., acceleration and gripping) and transient faults in robotic systems, which often

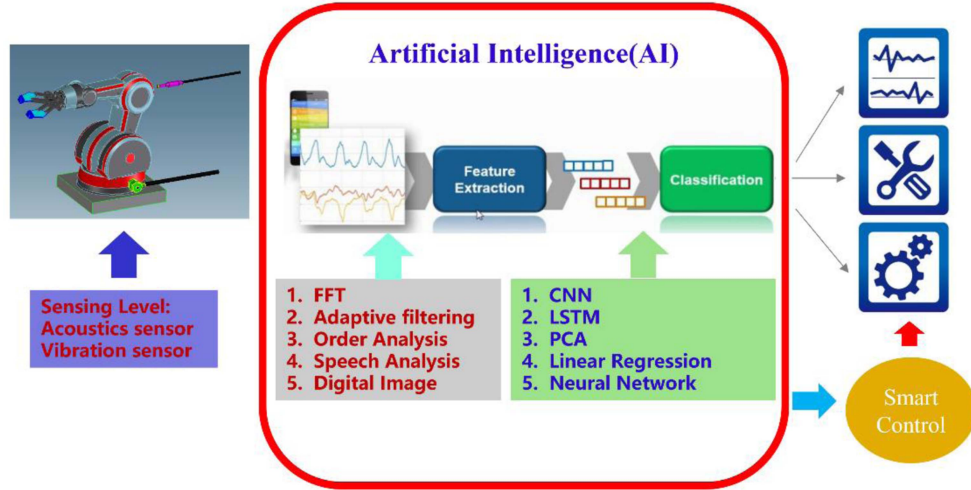


Fig. 2. AI-assisted fault diagnosis flowchart for a robotic arm.

exhibit nonstationary behaviors due to varying operational conditions [71].

1) *Short-Time Fourier Transform*: STFT applies the Fourier transform on a sliding time window, producing a spectrogram—a 2-D representation of how the signal's frequency content evolves over time.

- 1) *Strengths*: STFT is simple to implement and is compatible with CNNs for classification tasks, making it appropriate for real-time fault detection in robotic systems [72].
- 2) *Limitations*: The fixed window size in STFT causes a tradeoff between time and frequency resolution, which may limit its effectiveness in capturing transient events with varying durations [72].

Spectrograms derived from STFT have been employed in the detection of joint misalignments and actuator anomalies in collaborative robotic systems, providing visual insights into the temporal evolution of fault-related frequency components [73].

2) *Wavelet Transform*: The WT enables multiresolution analysis by projecting the signal onto scalable wavelets, allowing examination of both high-frequency transients and low-frequency trends [74].

- 1) *Continuous wavelet transform (CWT)*: CWT gives high-resolution time-scale representations, typically visualized as scalograms, which are effective in identifying transient features in vibration signals [74].
- 2) *Discrete wavelet transform (DWT)*: DWT promotes hierarchical decomposition of signals, is commonly used for denoising and compression, and is especially effective in detecting faults, including sharp transients or sudden changes in signal characteristics, such as joint impact or backlash [14], [71].

Wavelet-based methods are especially effective in detecting faults involving sharp transients or sudden changes in signal characteristics. For example, DWT has been successfully applied to diagnose backlash faults in industrial robot joints by extracting notable features from vibration signals and classifying them using an artificial neural network [71].

3) *Empirical Mode Decomposition*: EMD is an adaptive technique that splits the signal into IMF, each of which represents oscillatory modes inherent in the data.

- 1) *Hilbert–Huang Transform (HHT)*: The HHT combines EMD with Hilbert spectral analysis to extract instantaneous frequencies, which makes it appropriate for analyzing nonlinear and nonstationary signals commonly found in robotic applications [14].
- 2) *Limitations*: Mode mixing may occur in EMD, where a single IMF contains multiple frequency components. Recent variants such as ensemble EMD and complete ensemble EMD with adaptive noise have been developed to mitigate this issue [75].

Applications of EMD include detecting stiffness degradation and unstructured contact anomalies in robotic actuators, giving a powerful tool for the analysis of complex nonlinear vibration signals [75].

D. Advanced Acoustic Feature Representations

To improve classification accuracy in robotic fault detection, several perceptually inspired or machine-learning-oriented feature representations have been proposed. These representations are designed to capture the subtle nuances of acoustic signals more effectively, thereby improving fault detection in robotic systems.

1) *Mel-Frequency Cepstral Coefficient*: Derived from the response characteristics of the human auditory system, MFCC emphasizes perceptually relevant frequency bands and has been extensively used in deep-learning-based fault detection across various mechanical systems. For example, in wind turbine gearbox diagnostics, MFCC features extracted from vibration signals have proved high classification performance. This shows their effectiveness in capturing fault-related features [76]. MFCC has proven to be superior to outperform traditional features in detecting faults in industrial motors, helping to identify early signs of mechanical degradation. In addition, they have been used in

detecting faults in robotic actuators and joints, which makes a significant contribution to the accuracy of fault classification.

2) Log-Mel Spectrograms: Log-Mel spectrograms convert Mel-scaled spectrograms to logarithmic amplitude, improving feature separation for CNN-based models. This transformation improves the discriminative power of features, making them more suitable for DL applications in fault detection. Recent studies in automated machinery fault detection have demonstrated that Log-Mel spectrograms significantly improve fault classification accuracy, especially when coupled with DL algorithms such as CNNs and LSTM networks [77].

3) Constant-Q Transform: The constant-q transform (CQT) provides high-frequency resolution at low frequencies, making it appropriate for detecting slow-evolving harmonic content in robotic movements. By giving a logarithmic frequency scale, the CQT can more effectively capture the harmonic structure of the sound, helping to identify faults characterized by such patterns. The CQT has been successfully applied to the detection of faults in rotating machinery, where it revealed distinct patterns associated with early-stage gear defects [78], [79].

4) Wavelet Scattering Transform: Wavelet scattering transform (WST) extracts translation-invariant features while preserving local deformation sensitivity, which is quite beneficial for fault detection using shallow neural architectures. A study on bearing fault classification demonstrated WST-based features when used with ensemble machine learning algorithms. This method achieved over 99% classification accuracy and demonstrated strong robustness in capturing fault-related information [80]. In addition, WST has been used in combination with deep neural networks for real-time robotic fault diagnosis, demonstrating high accuracy in detecting sudden mechanical failures such as joint misalignments and actuator anomalies.

These advanced feature representations have been empirically shown to outperform traditional features in multiple robotic fault detection scenarios, especially when combined with DL frameworks. For example, a study on fault diagnosis in a four-arm delta robot utilized wavelet scattering networks and AI techniques to get high classification accuracy, emphasizing the importance of effective feature extraction and classification methods in complex robotic systems [76].

E. Denoising and Enhancement Techniques

Effective noise suppression is critical in real-world deployments, where robotic arms usually operate in acoustically cluttered environments. Noise can obscure fault-related acoustic signals, leading to decreased diagnostic accuracy. Implementing robust denoising and enhancement techniques is essential to improve the reliability of fault detection systems.

1) Spectral Subtraction: Spectral subtraction is a widely used technique that reduces stationary background noise by subtracting an estimated noise profile from the signal spectrum. This method is quite useful in environments where noise characteristics remain relatively constant over time. However, it may cause musical noise artifacts if not carefully implemented. Recent advancements have focused on adaptive spectral subtraction methods, which can dynamically adjust to changing

noise conditions, improving their effectiveness in real-world applications [81].

2) Wiener Filtering: Wiener filtering adapts to signal statistics for optimal noise suppression with minimal signal distortion. It operates by estimating the SNR at each frequency band and applying a gain function to suppress noise accordingly. This technique has been successfully applied to both speech enhancement and mechanical fault detection, effectively improving the SNR while keeping the integrity of the target signal [82].

3) Independent Component Analysis: Independent component analysis (ICA) is used to separate acoustic sources in multisensor setups, which improves the isolation of fault-related signals. By assuming that the observed signals are linear mixtures of non-Gaussian source signals, ICA can effectively separate them. This method has been employed in machine fault diagnosis to extract weak fault signals from noisy measurements, thereby improving detection accuracy. For example, the study demonstrated the application of ICA in bearing fault diagnosis, which successfully separated fault-related components from noise, thereby improving diagnostic performance [83].

4) Denoising Autoencoders: Using neural networks, denoising autoencoders (DAEs) are trained to transform noisy inputs into clean outputs by capturing the underlying structure of the data. They have become an increasingly common preprocessing technique in DL pipelines. DAEs have been used in various fields, including fault detection in analog circuits and rotating machinery. For example, an end-to-end DAE-based method was used for diagnosing faults in analog circuits, achieving high accuracy with minimal preprocessing requirements [84]. In a similar way, DAE-based models have also been applied to bearing fault diagnosis, effectively classifying fault types using raw time-domain vibration data [85].

These methods improve the quality of input features and the robustness of downstream classifiers. By effectively reducing noise while keeping important signal characteristics, they contribute to more accurate and reliable fault detection in robotic systems.

IV. FAULT CLASSIFICATION MODELS

Once meaningful features have been extracted from the acoustic signals, the next important step is fault classification—the process of assigning diagnostic labels (e.g., normal operation, gear wear, misalignment, and actuator failure) based on the learned patterns. Classification models in acoustic-based fault detection are roughly divided into traditional ML models, DL architectures, and hybrid or ensemble methods. Each paradigm offers distinct advantages depending on the complexity of features, volume of data, and deployment scenario. This section systematically reviews fault classification models with a focus on applications to robotic arms.

A. Traditional ML Models

Traditional ML models have been widely used in fault diagnosis due to their computational efficiency, ease of implementation, and interpretability. These models rely heavily on manually

extracted features and are usually constrained in adaptability across variable acoustic environments.

1) *Support Vector Machines*: SVMs are quite effective in high-dimensional feature spaces and are well suited for binary classification tasks. Their kernel-based approach, such as radial basis function, allows them to model nonlinear decision boundaries, making them suitable for distinguishing subtle spectral or temporal variations in acoustic signals [86].

- 1) *Pros*: They are robust to overfitting; they perform well with small datasets.
- 2) *Cons*: They require precise feature selection and hyperparameter tuning.

SVMs have been effectively used to classify motor bearing faults using MFCCs, with reported classification accuracy exceeding 95% in controlled acoustic settings [87].

2) *k-Nearest Neighbors*: The k-NN algorithm classifies input data by calculating the distance between it and its k -NN in the feature space. Its nonparametric nature and brevity make it an attractive option for lightweight real-time robotic applications.

- 1) *Pros*: It does not require explicit training; it is easy to implement.
- 2) *Cons*: It is sensitive to noise; its performance degrades in high-dimensional feature spaces.

k-NN has demonstrated moderate classification accuracy in robotic systems, especially in identifying subtle acoustic anomalies such as end-effector faults [88].

3) *Decision Trees and RFs*: Decision trees create classification rules based on feature thresholds, while RFs aggregate multiple trees to improve generalization and reduce overfitting [89]. These models are quite effective for heterogeneous acoustic feature sets that include both time- and frequency-domain characteristics.

- 1) *Pros*: Decision trees offer high interpretability; they are capable of modeling nonlinear relationships.
- 2) *Cons*: Decision trees may overfit; ensemble methods may lack transparency.

RFs have demonstrated strong performance in identifying joint degradation in lightweight robotic manipulators under variable conditions [90].

4) *Gaussian Mixture Models and Hidden Markov Models*: The GMM models data as a weighted sum of Gaussian distributions, while the HMM incorporates temporal dynamics through probabilistic state transitions. These models are highly appropriate for analyzing sequential acoustic patterns, for example, those produced by rotating components or motion sequences [91].

- 1) *Pros*: They are effective for nonstationary and time-evolving signals.
- 2) *Cons*: They require careful initialization and parameter tuning.

These models have been used in acoustic-based robotic control systems, such as voice-commanded robotic arms and sequence modeling of actuator behavior in dynamic states [92].

Despite their strengths, traditional ML models often face challenges in adapting to new robotic platforms or changing acoustic environments. This limitation has caused growing interest in DL models capable of learning robust features directly from raw or preprocessed signals.

B. DL Approaches

DL models overcome limitations of feature dependence by learning higher level representations directly from raw or transformed acoustic data. These methods are quite effective in diverse industrial fault detection tasks, especially in nonstationary acoustic environments.

1) *Convolutional Neural Networks*: CNNs have demonstrated significant success in fault diagnosis tasks, particularly when dealing with 2-D acoustic representations such as spectrograms, wavelet scalograms, and log-Mel spectrograms. By capturing intricate temporal and spectral patterns, these representations are quite effective in detecting mechanical faults in rotating machinery [93].

1) *Architecture*: CNNs usually consist of multiple convolutional layers with rectified linear unit (ReLU) activation functions, pooling layers, and dense classification layers. This architecture can make the network automatically learn hierarchical features from raw input data, eliminating the need for manual feature extraction [93].

2) *Applications*: CNNs have been widely used to detect mechanical faults such as gear pitting, joint wear, and actuator anomalies. For example, an end-to-end CNN model developed for gear fault diagnosis using sound signals is quite effective without the need for manual feature engineering [94]. Another application involved using CNNs with vibration and acoustic signals for more comprehensive fault detection in gears [95].

3) *Variants*: In order to solve specific problems and enhance performance, numerous CNN variants have been introduced.

- a) *ResNet*: It introduces residual connections to facilitate the training of deeper networks, improving performance on complex datasets [96].
- b) *DenseNet*: It employs dense connections between layers, promoting feature reuse and improving efficiency [97].
- c) *MobileNet*: MobileNet is designed for mobile and edge devices and utilizes depthwise separable convolutions to reduce computational complexity while maintaining accuracy [97].

These variants have been employed to enhance depth, reduce overfitting, and improve performance on edge devices, which makes them appropriate for real-time fault diagnosis applications in collaborative robots.

2) *RNNs, LSTM, and Gated Recurrent Unit*: RNNs enable the modeling of temporal patterns in acoustic signals by handling sequential data. However, due to the vanishing gradient problem, traditional RNNs usually have difficulty in handling long-term dependencies. LSTM and gated recurrent units (GRUs) solve this problem by using gating mechanisms to regulate the flow of information, which makes them suitable for modeling long-term dependencies in time-series data [98].

1) *Use case*: RNNs, LSTMs, and GRUs have been effectively applied in real-time acoustic fault tracking in robotic arms, especially in scenarios where joint behavior exhibits temporal variability. These models can

capture the dynamic changes in acoustic signals over time, thereby promoting the detection of faults that manifest through subtle temporal patterns [99].

- 2) The hybrid method combining CNNs with LSTMs has proven effective in systems with nonrepetitive motion profiles. CNNs are employed for feature extraction from raw acoustic signals, while LSTMs model the temporal dependencies of the extracted features. This hybrid method improves the model's ability to detect faults by using both spatial and temporal information [100].

3) *AEs and VAEs*: AEs are unsupervised neural networks that learn compressed representations of input data by minimizing reconstruction error. When trained exclusively on normal operational data, AEs can effectively detect anomalies by identifying deviations in reconstruction error, making them quite suitable for fault detection in scenarios where labeled fault data are scarce [101].

- 1) *Advantages*: In unsupervised fault detection tasks, AEs are quite effective when labeled data are limited. These models are capable of learning normal system behavior and detecting deviations without the need for explicit fault labels [102].
- 2) *Use case*: In industrial environments, AEs have been used for acoustic-based anomaly detection in robotic joints. Through the reconstruction of normal acoustic signals, AEs are able to identify unknown failure modes by detecting the increased reconstruction errors [103].

VAEs extend AEs by introducing a probabilistic latent space, thereby allowing the model to capture uncertainty in the data. This probabilistic modeling makes VAEs quite effective in noisy or highly variable acoustic environments [104].

- 1) *Advantages*: VAEs can better capture the basic distribution of normal operating data, thus enabling more robust anomaly detection under various conditions [105].
- 2) *Use case*: VAEs have been utilized for unsupervised fault diagnosis in industrial equipment and robotic systems, giving improved generalization and probabilistic interpretations of anomalies through reconstruction distributions and latent features [106].

Table I summarizes the advantages and disadvantages of the technology.

4) *Transformer-Based Models*: Transformers apply attention mechanisms to model global dependencies and temporal structures in acoustic sequences. Unlike traditional models, they can capture long-range interactions within the data, making them quite effective for complex acoustic modeling tasks.

- 1) *Architecture*: The AST, a recently developed architecture, is designed to directly handle audio spectrograms and has shown great performance in audio classification tasks [51]. By integrating CNNs with transformers, the Conformer model effectively captures both local and global representations in speech recognition tasks [107]. In addition, vision transformers have been adapted for time–frequency data and can effectively process spectrograms for acoustic event classification [108].
- 2) *Benefits*: Transformer-based models give excellent scalability and performance on long sequences, providing better generalization to irregular acoustic patterns. Their

self-attention mechanisms can effectively model complex dependencies in the data [109].

- 3) *Challenges*: Although transformers give significant advantages, transformers require large datasets and substantial computational resources for training, which can be a barrier to their widespread adoption in some applications [110].

Recent works have proved the effectiveness of pretrained transformer models fine-tuned for robotic acoustic diagnostics. For instance, the acoustic spectrogram transformer - sound event detection (AST-SED) model, based on the AST, has shown promising results in sound event detection tasks relevant to robotic applications [110].

C. Hybrid and Ensemble Models

Hybrid models combine multiple learning paradigms to improve classification accuracy and robustness, especially in complex acoustic environments encountered in robotic applications. These models use the strengths of different algorithms to address the limitations inherent in individual methods.

- 1) *CNN-LSTM*: The combination of CNNs with LSTM networks can extract spatial features and model temporal dependencies in acoustic signals at the same time. This architecture has proved high accuracy in classifying various defects and has been effectively applied to tasks like damage detection in metallic pipelines using ultrasonic guided Lamb waves [111]. Likewise, in the medical field, a CNN-LSTM hybrid model has been applied for automated lung sound classification, which demonstrates the model's versatility in handling complex acoustic patterns [112].
- 2) *Feature fusion*: Model performance can be improved by enriching the feature space through the combination of handmade features (e.g., RMS and kurtosis) and learned representations (e.g., MFCCs and CNN embeddings). For example, research has shown that combining MFCC features with deep features extracted by CNNs improves the accuracy of heart sound classification, emphasizing how feature fusion effectively captures both domain-specific and data-driven information [113]. In addition, time–frequency feature fusion techniques have been used to improve noise robustness in audio event classification, demonstrating their effectiveness in challenging acoustic environments [114].
- 3) *Ensemble methods*: They combine the outputs of multiple models to improve prediction accuracy and robustness. These techniques, such as voting or stacking, can be used to aggregate the forecasts of models, such as CNNs, RFs, SVMs, and AEs. Ensemble methods have proven effective in acoustic signal classification, especially for improving fault detection in mechanical systems by increasing robustness to noise and data variability [115]. Moreover, ensemble learning has been utilized in structural health monitoring applications, where combining different classifiers has led to improved damage detection performance in complex structures [116].

TABLE I
STRENGTHS AND WEAKNESSES OF TECHNOLOGIES FOR ROBOTIC ARM FAULT DIAGNOSIS

Technology	Specific technique	Advantages	Disadvantages
Model-based	Physical models, state observers, power consumption models [4]	High interpretability: Models are based on the system's physical principles, making fault causes understandable [1]. Early detection: Sensitive to small deviations from model predictions, enabling incipient fault detection [2]. No need for large datasets: Does not rely on extensive historical fault data once the model is established.	Difficult model development: Building accurate mathematical models for complex nonlinear systems, nonlinear system, such as robotic arms, is a difficult and time-consuming challenge [2]. Poor adaptability: A fixed model may become inaccurate under varying operating conditions (e.g., load, speed) or due to system aging, thereby causing false alarms [16]. Limited to known faults: Models are generally designed for specific, anticipated faults and have poor performance on unmodeled or novel failures [1].
Signal processing & analysis	Vibration analysis [8], [9], [62] AE [5], [6], [7] Audible sound analysis [37], [42]	High sensitivity: Considered the gold standard in mechanical diagnosis, it demonstrates high sensitivity to defects in rotating components such as bearings and gears [7], [65]. Mature technology: Supported by mature theory and analysis tools (e.g., FFT and envelope analysis) [36]. Excellent early detection: Which can detect the transient stress waves from microscopic crack initiation and other early faults [40]. Location-independent: A single sensor can monitor a large area; it is not as sensitive to precise placement as a vibration sensor [5]. Effective for low speeds: Performs well in situations where the vibration signal is weak (like low-speed rotating machinery) [7]. Noncontact and Low cost: Enables remote and noninvasive monitoring by using inexpensive microphones [47]. Simple installation: Flexible deployment, as sensors do not need to be mounted directly on the robot's structure [105].	Contact-based measurement: Requires physical mounting of the sensor (accelerometer), which may be affected by placement and coupling [9]. Complex transmission path: The transmission path from the fault source to the sensor may introduce distortions, thereby complicating analysis [62]. Susceptible to ambient noise: It exhibits high sensitivity to high-frequency environmental noise (e.g., pneumatics and metal impacts), which may obscure or mask potential fault signals [6], [39]. Large data volume: AE signals are high-frequency and continuous, generating large amounts of data that require significant storage and processing power. Low SNR: In industrial environments, target fault signals can easily be drowned out by the complex and changing background noise [41, 105]. Affected by acoustics: Signal quality and diagnostic stability are severely influenced by environmental factors, such as reflection, attenuation, and interference.
Data-driven /AI	Traditional ML (SVM, RF, GMM, and HMM) [17], [48], [49], [50], [88]	No physical model required: It enables the model to learn patterns directly from data without requiring prior knowledge of the system's physical principles [17]. Computationally efficient: In contrast to DL, it can be trained faster on small- and medium-sized datasets and does not require many computational resources [87].	Reliance on manual feature engineering: Performance is severely dependent on handmade features designed by domain experts, a time-consuming and skill-intensive process [13], [44]. Limited generalization: Model performance can degrade obviously under changing operating conditions, requiring feature reengineering and retraining [16].

TABLE I
(CONTINUED.)

Technology	Specific technique	Advantages	Disadvantages
	DL (CNN, RNN/LSTM, AE and Transformer) [13], [18], [52]	Good performance on specific tasks: It shows good performance on classification tasks where the input features are well defined and highly discriminative [90]. Automatic feature learning: By autonomously learning hierarchical and abstract features from raw data (e.g., time series and spectrograms), it eliminates the need for manual feature design [3], [93]. Powerful nonlinear modeling: It is able to model complex high-dimensional data and effectively achieve high accuracy in fault diagnosis [97], [98]. Excellent transferability: Transfer Learning enables models trained in one domain to be effectively adapted to another, thus eliminating reliance on large labeled data in the target domain [23], [26], [60].	Requires large datasets: A large amount of labeled data are usually required for training, which are often lacking in industrial settings [13], [16]. High computational cost: Training deep models is computationally intensive, usually requiring powerful hardware (e.g., GPUs) and long training times [44]. Poor interpretability (black box): It is difficult to understand the reasoning behind a model's diagnostic decisions, limiting trust and adoption in high-stakes applications [13].

Table II summarizes the algorithms. These hybrid and ensemble strategies are quite effective in robotic environments characterized by motion diversity, acoustic interference, and hardware heterogeneity, giving robust solutions for acoustic-based fault detection and diagnostics.

D. Performance Evaluation Metrics

The effectiveness of fault classification models is evaluated through multiple criteria: accuracy, precision, recall, and F1-score. These are basic classification metrics used to evaluate model accuracy. Accuracy reflects the overall correctness of the model, whereas precision and recall give more detailed insights into the model's performance on identifying positive class predictions. The F1-score, being the harmonic mean of precision and recall, offers a balance between the two, especially useful in imbalanced datasets. These metrics are quite important in edge computing scenarios where real-time fault detection is critical, such as in IIoT applications [117].

- 1) *Confusion matrix:* This tool reveals classwise performance and common misclassification patterns, enabling a deeper understanding of model behavior. It is quite effective when dealing with multiple fault categories, as it highlights specific areas where the model may not be good.
- 2) *Receiver operating characteristic (ROC) curve and area under the curve (AUC):* The ROC curve evaluates the discriminative performance of the model at different threshold settings, while the AUC gives a single scalar value to summarize the model's ability to distinguish between classes. These metrics are valuable for comparing different models or configurations in fault detection tasks.
- 3) *Latency and computational load:* These metrics are quite important for edge deployment on robotic platforms with

limited resources. The time required for inference and the computational resources consumed are evaluated to ensure that the model fits within the immediate constraints of such systems. Studies have emphasized the importance of these factors in deploying effective fault detection systems in resource-constrained environments [117].

In practical applications, models trained under controlled laboratory conditions usually experience noticeable performance degradation when exposed to unmodeled factors such as ambient noise, fluctuating payloads, or microphone misalignment. This phenomenon, commonly referred to as domain shift, presents substantial obstacles to the successful deployment of models in real-world applications. To address this, robustness, domain adaptation, and transfer learning have become critical areas of research. For instance, domain-adversarial neural networks have been proposed to improve model performance on target domains by minimizing the discrepancy between source and target data distributions [23]. Several methods have demonstrated effectiveness in improving the generalization ability of fault diagnosis models under different operational conditions.

In addition, integrating expert knowledge with domain adaptation techniques has been explored to overcome the limitations posed by the scarcity of labeled fault data in real-world settings. By generating synthetic fault data and adapting models trained on these data to practical applications, researchers have obtained higher fault diagnosis performance [25]. Another study emphasized the significance of accounting for environmental noise in fault diagnosis. To solve this problem, unsupervised learning methods combined with domain adaptation strategies were employed to develop models capable of maintaining high performance under noisy operational conditions [117].

Finally, deep transfer learning (DTL) has emerged as a powerful tool to improve the robustness of fault diagnosis models. By using knowledge from related domains, DTL promotes the

TABLE II
ACOUSTIC ALGORITHMS FOR ROBOTIC ARM APPLICATIONS

Type	Algorithm/technique	Description and application
Preprocessing and denoising	WPT	Description: A powerful time–frequency analysis tool that decomposes a signal into distinct frequency bands, thereby effectively separating noise from useful information [72], [74]. Application: It can be used to suppress background noise generated by robotic arm operations or to isolate transient components in the acoustic signal that are indicative of specific faults, such as bearing defects or mechanical clearance issues [71], [73], [115].
	EMD/VMD	Description: Adaptive signal decomposition methods that break down complex, nonlinear, and nonstationary signals into a series of IMFs, suitable for analyzing complex robotic arm acoustics [14], [15]. Application: Decomposes the raw sound signal to highlight specific oscillatory modes associated with faults for subsequent feature extraction and analysis.
	DAE	Description: An unsupervised neural network that learns to reconstruct a clean original signal from a noisy input [14], [81]. Application: Commonly employed as a preprocessing step to clean acoustic signals before they are fed into a diagnostic model, it improves signal quality by removing background and random industrial noise, thereby improving the SNR and boosting the accuracy of subsequent fault classifiers [85], [105].
Feature extraction	Time-domain statistical features	Description: Statistical features, such as RMS, Kurtosis, Skewness, and Crest Factor, are computed directly from the original signal waveform to characterize its amplitude and distributional properties [43], [66]. Application: provides a simple and fast method for initial anomaly detection. For instance, an increase in RMS can indicate higher loads or wear, while a high Kurtosis value is often associated with impulsive faults such as bearing spalls [63].
	Frequency-domain features	Description: The signal is converted from the time domain to the frequency domain by using the FFT to analyze its PSD [68]. Application: Used to identify characteristic fault frequencies and their harmonics from components such as motors and gear reducers in the robot's acoustic signature, thereby helping to diagnose issues such as imbalance, misalignment, or gear wear [69], [93].
	time–frequency/ spectrogram features	Description: Several methods, such as STFT, MFCCs and Spectrograms, can visualize how frequency content evolves [45], [70]. Application: Converts 1-D sound signals into 2-D image-like representations (spectrograms). These are ideal inputs for CNNs, allowing them to learn time-varying fault patterns as visual features (21, 56, 113). The Kurtogram can be used to identify the optimal frequency band for demodulation to highlight impacts [35], [94].
	WST	Description: A deep feature extractor that builds stable and invariant features through cascaded WTs and modulus operators. It is robust to signal translations and minor deformations [76]. Application: As a robust feature extractor, it is quite suitable for robots operating under variable conditions (speed and load). It enhances model generalization by providing stable acoustic features [76], [77], [78].

TABLE II
(CONTINUED.)

Type	Algorithm/technique	Description and application
Classification & Diagnosis	CNN	Description: A DL model that excels at processing grid-like data, such as images [13]. Application: The main model for fault diagnosis by using acoustic signals. It takes spectrograms as input and automatically learns to identify visual patterns corresponding to different fault types in bearings, gears, etc. [93, 95, 96].
	RNN / LSTM	Description: A class of neural networks adept at modeling sequential data and capturing temporal dependencies [18]. Application: Analyzes the temporal evolution of acoustic signals. It is usually combined with CNNs in hybrid architectures (e.g., CNN-LSTM), where the CNN extracts spatial (spectral) features and the LSTM models their temporal relationships, boosting diagnostic performance [53, 97, 98].
	AE / VAE	Description: Unsupervised models that learn to reconstruct their input. They detect anomalies based on reconstruction error: low error for normal data, high error for anomalous data [101, 106]. Application: Particularly effective in anomaly detection when only healthy data are available. An AE/VAE trained by using normal robot sounds can then detect any unknown deviation, signaling a potential fault without needing prior fault samples [59, 102, 103].
	Transformer Models	Description: Models based on a self-attention mechanism that captures long-range dependencies in sequential data, which first appeared in natural language processing [107]. Application: Models like the AST treat a spectrogram as a sequence of patches and use attention to learn global relationships between spectral features. They show great promise for robust sound event detection and fault diagnosis [51, 55, 109].

development of models capable of adapting to novel and previously unseen conditions, even when labeled data are scarce, thereby addressing the challenges posed by domain shifts and limited data availability [118].

V. EDGE INTELLIGENCE DEPLOYMENT AND IMPLEMENTATION FOR ACOUSTIC DIAGNOSIS OF INDUSTRIAL ROBOT ARMS

While DL models have demonstrated exceptional accuracy in acoustic fault diagnosis tasks, successfully transferring these complex models from academic research (or cloud environments) to resource-constrained time-critical industrial settings still presents significant practical challenges. Monitoring robotic arms within an Industry 4.0 environment not only demands high-precision low-latency inference feedback but also necessitates careful consideration of deployment cost, power consumption limits, and seamless integration with existing IIoT systems.

To bridge the gap between theoretical modeling and industrial realization, edge intelligence is established as an indispensable strategic element. This section aims to provide an in-depth exploration of the key factors required to achieve efficient, reliable, and scalable acoustic monitoring systems for robotic arms. The discussion is structured across the following three tiers.

- 1) *Quantitative performance evaluation:* Initially, the performance, latency, and energy efficiency of typical edge hardware (ranging from microcontrollers (MCUs) to AI

accelerators) executing inference will be quantitatively analyzed based on established public benchmarks.

- 2) *IIoT deployment architecture:* Next, the discussion will address how to select appropriate hardware within the tiered architecture of the IIoT (from the extreme edge to the edge gateway) and consider the design tradeoffs involved in real-time data streaming and model compression strategies.
- 3) *Industrial implementation and commercial solutions:* Finally, this section will summarize current commercialized solutions and industrial case studies, illustrating how acoustic monitoring technology is practically applied to quality control and predictive maintenance, thereby validating the commercial viability and practical value of Edge AI deployment.

A. Edge Deployment and Quantitative Performance Evaluation

This section compiles a comparative analysis of performance metrics for three representative tiers of edge hardware—MCUs, single-board computers (SBC), and AI accelerators—utilizing publicly measured data. The evaluation draws upon recognized public benchmarks, including the MLPerf Inference Benchmark (v2.1–v3.0), the NVIDIA TensorRT Performance Report, official Google Coral testing data, and the STMicroelectronics

TABLE III
PERFORMANCE AND COST ANALYSIS OF EDGE AI PLATFORMS

	NVIDIA Jetson Orin Nano (8GB)	Raspberry Pi 5 (8GB)	STM32H7 (Example: NUCLEO-H7A3ZI-Q)
Architecture	1024-core Ampere GPU, 32 Tensor Cores, 6-core ARM CPU [120]	Quad-core ARM Cortex-A76 [123]	32-bit ARM Cortex-M7 [124]
AI Performance	40 TOPS (Sparse INT8) [120]	N/A (No Dedicated Accelerator)	N/A (Micro-ops/cycle)
Power Consumption	7 – 15W [120]	~5W (Typical Load)	< 1W (e.g., STM32U3 KWS at 245mW) [122]
Development Cost (Kit)	~\$249 [125]	~\$80 (Board Only) [126]	~\$100 [127]
Inference (ResNet-50) Latency	~0.64 ms (On AGX Orin) [119]	~423 ms/image [121]	Not Applicable (Model too large)
Inference Latency (CNN)	Estimated < 5 ms	~108 ms/image (SqueezeNet) [121]	~20.7 ms (For Keyword Spotting Model) [122]
Computational Cost	High (Suitable for complex, parallelizable models)	Medium (CPU-dependent; unsuitable for heavy parallel processing)	Extremely Low (Applicable only to highly optimized TinyML models)
Deployment Feasibility	Complex CNN/Transformer Real-Time Inference	Prototyping, Medium Complexity Models	Highly Optimized, Low-Power TinyML Models

TinyML Benchmark. The performance indicators encompassed within this review are: 1) inference latency; 2) model size and parameter count; 3) computational cost; and 4) energy efficiency tradeoff across different hardware tiers (deployment feasibility). The resulting comparative data are summarized in Table III.

Through strategic model optimization techniques, such as knowledge distillation and quantization, lightweight edge models are capable of achieving low inference latency while maintaining high accuracy. This is evidenced by the reduction in ResNet-50 inference latency, which can drop to as low as 0.64 ms on high-end edge platforms (e.g., NVIDIA Jetson AGX Orin) or remain at approximately 423 ms on platforms such as the Raspberry Pi 5. This performance characteristic directly satisfies the need for real-time intervention required in industrial automation. Furthermore, the substantial reduction in model complexity, exemplified by an 83% decrease in floating point operations (FLOPs), is deemed critical for enabling large-scale cost-effective Internet of Things (IoT) deployment.

Recent studies confirm that the application of model compression, quantization, and knowledge distillation allows DL models to achieve significant performance improvements, enabling them to meet industrial real-time requirements. For instance, edge-optimized models can realize over 96.37% compression in parameter count and an 83.08% reduction in computational load (FLOPs) compared to their traditional cloud-based counterparts [128]. This extensive resource reduction critically lowers the demand for GPU memory, thereby facilitating deployment on lower cost edge devices. In practical operational terms, this optimization translates directly into lower latency.

Empirical data demonstrate that the average inference time for lightweight edge models can be reduced to 24.0 ms, nearly half the 47.2 ms recorded for cloud models [128], representing an average overall inference speed improvement of 80.47%. In high-speed industrial robotics scenarios, the timeliness of the diagnostic result is paramount, as real-time diagnosis must be completed within the cycle time of the entire process to enable immediate control feedback. Consequently, low latency often constitutes a more critical design constraint than marginal accuracy gains.

B. IIoT Architecture and Industrial Deployment

This section explores the tiered architecture of edge devices—specifically the sensor node, gateway, and cloud—and their associated communication protocols [e.g., message queuing telemetry transport (MQTT) and constrained application protocol (CoAP)]. It discusses the crucial design considerations concerning latency and energy efficiency when deploying acoustic diagnostic systems within an IIoT environment. Model deployment is generally segmented into three principal layers, the comparison of which is summarized using specific technological data in Table IV.

1) *Cloud*: It is designated for complex model training without strict real-time requirements, massive data storage, and advanced processes, such as knowledge distillation [128].

2) *Edge Server/Single-Board Computer (SBC/Edge Gateway)*: Platforms, such as the NVIDIA Jetson Xavier NX/Orin

TABLE IV
PERFORMANCE COMPARISON OF COMPUTATIONAL PLATFORMS FOR MODELS

Model/Platform Classification	Representative Hardware (Example)	Parameter Count (Params) Reduction	Inference Latency	Computational Load (FLOPs) Reduction	Real-Time Advantage
Cloud Large-Scale DL (Cloud Model)	Data Center-Grade GPU	N/A	47.2 m [128]	N/A	High-Accuracy Training
Edge-Optimized DL (SBC/Jetson)	NVIDIA Jetson Xavier NX	96.37% Compression	24.0 ms [128]	83.08%	Meets Real-Time FDD Requirements (80.47% Speed Improvement)
Extreme Edge (TinyML)	MCU (with NPU) [132]	Maximally Reduced [130]	Millisecond Range [131]	Extremely Low (Ultra-Energy Efficient)	Ultra Low Power Monitoring

series [129], fall into this category. These systems are equipped with neural processing unit (NPU)/GPU accelerators, making them suitable for executing quantized and optimized DL models, including variants of ResNet or Transformer [129]. They possess the capability to process complex features, such as 2-D spectrograms, thereby meeting real-time demands (with latencies around 24 ms [128]).

3) *Extreme Edge/MCU*: These are specifically engineered for ultralow power consumption and minimal latency, running TinyML solutions [130]. Such platforms typically perform simple time-domain feature extraction and preliminary anomaly assessment. Their power consumption and physical dimensions are critical factors for deployment in severely resource-constrained locations, such as a robot's end-effector [131].

Furthermore, edge computing hardware can be broadly categorized into two main classes: Extreme Edge/Microcontroller (MCU/TinyML) [130] and Edge Server/Single-Board Computer (SBC/Jetson) [119].

- 1) *MCU platforms (TinyML)*: These are ideal for running extremely lightweight low-power CNN models, designated for initial real-time anomaly screening, for instance, on battery-powered robotic components [132].
- 2) *SBC platforms (Jetson/NPU)*: Equipped with GPU/NPU acceleration, they are capable of handling compute-intensive tasks, such as transforming acoustic signals into 2-D spectrograms and subsequently executing optimized larger CNN networks or various Transformer models [129].

This analysis is intended to provide actionable insights, guiding engineers in selecting the appropriate edge computing module hardware based on the robotic arm's immediate operational constraints (e.g., travel speed, runtime, and power limitations) and the complexity of the required model (e.g., hybrid CNN-LSTM models or lightweight Transformer variants), ultimately facilitating efficient and sustainable edge IIoT deployment.

C. Industrial Deployment and Commercialization of Robotic Arm Acoustic Monitoring

In recent years, AI coupled with acoustic analysis techniques has been increasingly applied to industrial automation and robotic monitoring. In contrast to the past, where AI algorithms were often confined to the research stage, numerous commercial solutions are now available that can effectively execute acoustic monitoring and anomaly detection in industrial settings. This trend signifies the accelerated adoption of DL and ML technologies within smart manufacturing. The following four distinct solutions are introduced, with their comparative information consolidated in Table V.

1) *Microflown Technologies*: The robotic acoustic scanner system by Microflown Technologies exemplifies the integration of acoustic measurement with a robotic platform. This system combines the proprietary Scan&Paint 3D/Acoustic Shape platform with Universal Robots (UR) collaborative robot arms. By placing the acoustic probe directly onto the UR, the system performs fully automated 3-D sound field scanning and noise source localization in hazardous or constrained industrial environments. The measurement path is definable via the UR programming touchscreen interface, and the use of the robot's built-in kinematics provides position data, thereby eliminating the need for external tracking systems. This ensures highly repeatable and traceable measurement quality. This solution highlights the potential of the robotic arm to serve as a high-precision sensor carrier in acoustic monitoring applications [133].

2) *TMA Innovation*: TMA Innovation's Defect Detection System by Sound represents a shift in acoustic monitoring from "passive observation" toward "active control." The primary input to the system consists of high-fidelity acoustic signals collected by a precision microphone mounted on the robotic arm. AI algorithms then analyze the acoustic signatures to determine

TABLE V
COMPARISON OF INDUSTRIAL ROBOT ACOUSTIC MONITORING SYSTEMS AND COMMERCIAL SOLUTIONS

System / Vendor	Primary Function (s)	Acoustic Technology Focus	Deployment Platform (IIoT)	Practical Advantages
Microflow Scan&Paint 3D [133]	3D Sound Field Visualization / Noise Source Localization	High-Precision Acoustic Probes	Collaborative Robots (UR series)	Non-Contact, High Repeatability, Applicable to Hazardous or Confined Environments
TMA Innovation System [134]	AI-Driven Real-Time Defect Detection	Proprietary AI Algorithm, Precision Microphone	Integrated Robot Control System	Real-Time Analysis and Automated Response Capability
Neuron Soundware Acoustic Analysis System [135]	Acoustic Signature Analysis and Fault Detection	Acoustic AI Models and Edge Computing Technology	Industrial Robots and Manufacturing Production Lines	Reduced Maintenance Costs and Enhanced Operational Efficiency
XMPro iBOS Predictive Maintenance Solution for Robotic Arms [136]	Real-Time Monitoring and Predictive Maintenance of Robotic Arms	IoT Sensor Monitoring of Vibration, Temperature, Current/Voltage, Acoustics/Network Integrity + ML and Digital Twin Models	Robotic Arm Production Lines for Automotive Assembly, Welding, and Painting	Enhanced Robotic Arm Reliability, Extended Lifespan, and Reduced Maintenance Costs

normalcy or fault conditions. In one notable application, the system achieved a 99% defect-free rate in a valve tightening process by detecting anomalies in real time and automatically triggering the robotic arm to remove the defective item. This demonstrates the practical utility of noninvasive, online, and real-time quality control [134].

3) *Neuron Soundware*: Neuron Soundware proposes an acoustic monitoring solution that integrates edge computing with AI, focusing on the continuous surveillance of industrial robots. Its platform continuously collects data via acoustic vibration sensors installed on various robot axes. ML inference is performed locally on the nEdge edge device, with results presented on the nGuard portal or integrated into the client's SCADA system. Immediate alerts are issued to maintenance teams upon anomaly detection. It is reported that their technology has yielded significant industrial benefits, including an approximate 25% increase in manufacturer productivity, a potential 60% reduction in failures, a 35% reduction in manufacturing costs, and a 5% decrease in energy consumption for the overall production process [135].

4) *XMPro*: XMPro's predictive maintenance solution for automotive robotic arms incorporates acoustic data alongside other sensor values, such as current, voltage, temperature, vibration, and network integrity, into a unified data stream. Utilizing a Data Stream Designer, the system combines mechanical/electrical and maintenance context data to calculate the Overall Equipment Effectiveness (OEE) for individual robots. Coupled with digital twin technology for scenario simulation,

AI performs anomaly detection, Remaining Useful Life estimation, and fault classification. When indicators or models assess a risk, the system automatically triggers alerts, generates work requests, and suggests appropriate actions. This process is supported by detailed analysis and real-time decision support provided via an interactive dashboard, forming an actionable closed-loop maintenance system [136].

Furthermore, recent research has indicated that incorporating explainability modules into AI models for predictive maintenance contributes to enhanced system transparency and trustworthiness. The study by Kummari [137] on automotive manufacturing robots emphasized the criticality of explainability for predictive maintenance decisions. Practically, this can be achieved by integrating attribution methods, such as Shapley Value, to assist in improving the comprehensibility and reliability of the maintenance decision-making process.

In summation, current commercial acoustic monitoring technology is progressively maturing and has been demonstrably applied to robotic arms and manufacturing lines. These case studies, particularly the specific metrics disclosed in [134] and [135], strongly confirm that AI-based monitoring has achieved quantifiable “factory-floor benchmarks” in terms of “performance” (e.g., 25% productivity increase) and “reliability” (e.g., 99% defect-free rate).

However, while cataloging these successes, the present study identifies a critical characteristic of current commercial practice: information opacity. This phenomenon is primarily attributable

to the fact that relevant technical details are frequently treated as commercially sensitive information, posing two primary barriers to the further advancement of the industry.

- 1) *Absence of a transparent “cost benchmark”*: Although certain reports, such as [135], mention a 35% reduction in costs following adoption, the widespread absence of data concerning the system’s deployment cost and total cost of ownership prevents potential adopters from conducting accurate return on investment assessments. This inhibits the broader dissemination of the technology.
- 2) *Ambiguity of core “technical details”*: Commercial solutions are often presented using generalized terminology, such as “AI analysis” [134] or “AI platform” [135], [136]. The lack of specificity regarding which particular academic algorithms (e.g., those based on CNN, AE, or Transformer) have genuinely overcome on-site challenges and been validated as “mature” solutions severs the practical feedback loop between academia and industry. This obstructs academic research from conducting deeper optimization efforts on models already field-proven by the industry.

Given these observations, it is posited by the present study that the critical future trajectory for this domain lies not only in algorithmic innovation but also in bridging the aforementioned information gap. This necessity aligns precisely with the need for explainability highlighted by Kummari [137]. A more transparent and explainable AI framework is expected not only to enhance decision-making trust but, in the long term, also to facilitate a balance between protecting intellectual property and promoting the standardization of industrial technology, thereby accelerating the comprehensive implementation of smart manufacturing and predictive maintenance.

VI. CONCLUSION

This review emphasizes the growing importance and potential of acoustic-based fault detection techniques in improving the operational reliability of robotic arms. The noncontact nature of acoustic sensing and its ability to instantly detect subtle anomalies render it a valuable diagnostic tool in automated and high-precision industrial environments. With the development of advanced signal processing techniques, such as WTs, time–frequency representations, and data-driven denoising, extracting discriminative features from complex acoustic signals has become increasingly feasible. When integrated with ML and DL models, these features can do accurate and early fault classification, even under varying noise conditions. Although these obvious advances have been made, the challenges of dealing with real-world acoustic data remain, including the lack of standardized datasets, the sensitivity of models to operational variations, and the complexity of deploying real-time solutions. It is therefore required to not only ensure model generalizability but also adapt to varying operational and working conditions.

Future research should focus on developing robust, adaptive, and interpretable models that can be deployed in practical settings, as well as on creating open-source datasets to

promote benchmarking and reproducibility. In conclusion, acoustic-based fault detection represents a promising method of integrating multiple sensors placed for building resilient and intelligent robotic systems in the era of smart manufacturing and other consumer applications. It provides a viable path toward intelligent and self-monitoring robotic systems in Industry 4.0 environments. By enabling efficient real-time analysis on the device side, acoustic-sensing-based fault diagnosis technology will provide industrial equipment with accurate predictive maintenance capabilities, thereby reducing unplanned downtime, extending equipment life, and minimizing resource waste, embodying a practical and sustainable solution.

Recent research indicates that DL models can significantly enhance feature discriminability and stability under varying motion trajectories, particularly in low SNR acoustic environments [138]. Simultaneously, fault diagnosis models integrated with XAI have been proven to augment decision-making transparency and security [139]. These findings collectively demonstrate that effectively managing nonstationary acoustic features and addressing domain transfer issues constitute one of the core challenges in developing intelligent robotic arm diagnosis systems.

Looking forward, the integration of edge-computing-driven acoustic fault diagnosis systems with Digital Twin platforms, such as NVIDIA Omniverse, holds promise for achieving real-time anomaly prediction and swift fault remediation. By utilizing on-device inference to mitigate communication latency and leveraging cloud-based simulation to support cross-platform deployment and model adaptation [140], it is possible to establish a closed-loop smart maintenance management mechanism, thereby promoting the sustainable and resilient development of Industry 4.0 manufacturing environments.

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Cihun-Siyong Gong (Senior Member, IEEE) received the B.S. degree in civil engineering from National Chiao Tung University (NCTU), Taiwan, in 2002, the M.S. degree in biomedical engineering from National Yang Ming University (NYMU), Taiwan, in 2005, and the Ph.D. degree in electrical engineering from National Central University (NCU), Taiwan, in 2008.

He has been immersed in the field of biomedical circuits and systems since the outset of his academic journey. Since 2009, he has been with the Industrial Technology Research Institute, Hsinchu, Taiwan. In 2013, he was an Adjunct Faculty Member with National Taiwan Normal University, Taipei, Taiwan, and National United University, Miaoli, Taiwan, before joining Chang Gung University, Taoyuan, Taiwan, where he was promoted to a Full Professor. During this time, he was also an Associate Researcher and a Principal Investigator with Chang Gung Memorial Hospital, Taoyuan, fostering strong collaborations with clinicians. From 2015 to 2016, he was a Visiting Scholar with the University of Southampton, Southampton, U.K. He is currently an Associate Professor with the Department of Electrical Engineering, National Central University, Taoyuan. His research interests include controlled electrical/electronic design for biomedicine and health care, novel learning-based rehabilitation engineering, smart clinical guiding systems, and artificial-intelligence-based vehicular technology.

Dr. Gong has received many honors for his technical work, including the 2008 NCU Outstanding Graduate Student Award, the 2009 Best Design Award from the Chip Implementation Center, and the 2009 Taipei International Invention Show and Technomart Gold Award. In addition to his academic publications, he holds 12 patents, which reflect his contributions to the field. He is an Editor for SCI-indexed journals, such as IEEE ACCESS, PLOS ONE, and Micromachines.



Chih-Hui Simon Su was born in Yunlin, Taiwan, in 1976. He received the bachelor's degree in the department of industrial education from the National Changhua University of Education, Changhua, Taiwan, in 1999, the master's degree in the department of vehicle engineering from Da-Yeh University, Changhua, in 2002, the Ph.D. degree in the department of mechanical engineering from National Chiao-Tung University in 2006, and the Ph.D. degree in the graduate institute of electrical engineering from

Chang Gung University, Taoyuan, Taiwan, in 2024.

He applied online dynamic signals to fluid-dynamic-bearing (FDB) behavior analysis, leading a new product development plan—cooling and heat dissipation fan of FDB (2006, Ministry of Economic Affairs, Taiwan). He has more than ten papers and patent copyright publications, both international and national. In 2024, he joined the Department of Industrial Electronic Engineering, College of Engineering, National Chin-Yi University of Technology, Taichung, Taiwan, as an Adjunct Assistant Professor. He is currently a Postdoctoral Researcher with the Department of Electrical Engineering, National Central University, Taoyuan, under the supervision of Prof. Cihun-Siyong Gong. His current research interests include use of machine learning algorithms for prognostic and health management, which includes the analysis of order tracking, pattern recognition, and real-time acoustic signals for automotive engines or machine rotors.

Dr. Su received the outstanding paper awards in 26th National Conference on Vehicle Engineering and the oral paper awards in the Annual Meeting of the Taiwanese Society of Biomedical Engineering 2021. His paper "Real Time Monitoring System for Automotive Tire Set Using an Acoustic Signal" for International Conference on Vehicle Technology and Intelligent Transport Systems 2016 was a candidate for the best paper award.



Kuo-Wei Chao received the B.S. degree in mechanical engineering from National Cheng Kung University, Tainan, Taiwan, in 2022, and the M.S. degree in biomedical engineering from National Yang Ming Chiao Tung University, Taipei, Taiwan, in 2024.

He is currently a Research Assistant with the Department of Electrical Engineering, National Central University, Taoyuan, Taiwan, under the supervision of Prof. Cihun-Siyong Gong. His research interests include artificial intelligence,

signal processing, and optical microphone technology.



Cheng-Yen Lin was born in Taichung, Taiwan, in 2002. He received the B.S. degree in vehicle engineering from the National Taipei University of Technology, Taipei, Taiwan, in 2024. He is currently working toward the M.S. degree with the Department of Electrical Engineering, National Central University, Taoyuan, Taiwan, under the supervision of Prof. Cihun-Siyong Gong.



Yu-Hua Chen was born in Taipei, Taiwan, in 2000. She received the B.S. and M.S. degrees in electrical engineering from Chang Gung University, Taoyuan, Taiwan, in 2022 and 2024, respectively. She is currently working toward the Ph.D. degree with the Department of Electrical Engineering, National Central University, Taoyuan, Taiwan, under the supervision of Prof. Cihun-Siyong Gong.

Her research interests include the development of machine learning algorithms for vehicular acoustic diagnostics, model optimization, and fault detection systems.

Dr. Chen received the Outstanding Paper Award at 26th National Conference on Vehicle Engineering in 2021. She has also participated in several research projects funded by the Ministry of Science and Technology, Taiwan, focusing on smart early warning systems for vehicle fault prediction.